

## Section 3: Organic chemistry

- a) Introduction
- b) Alkanes
- c) Alkenes
- d) Ethanol

### a) Introduction

*Students will be assessed on their ability to:*

- 3.1 explain the terms homologous series, hydrocarbon, saturated, unsaturated, general formula and isomerism.

### b) Alkanes

*Students will be assessed on their ability to:*

- 3.2 recall that alkanes have the general formula  $C_nH_{2n+2}$
- 3.3 draw displayed formulae for alkanes with up to five carbon atoms in a molecule, and name the straight-chain isomers
- 3.4 recall the products of the complete and incomplete combustion of alkanes
- 3.5 recall the reaction of methane with bromine to form bromomethane in the presence of UV light.

### c) Alkenes

*Students will be assessed on their ability to:*

- 3.6 recall that alkenes have the general formula  $C_nH_{2n}$
- 3.7 draw displayed formulae for alkenes with up to four carbon atoms in a molecule, and name the straight-chain isomers
- 3.8 describe the addition reaction of alkenes with bromine, including the decolourising of bromine water as a test for alkenes.

### d) Ethanol

*Students will be assessed on their ability to:*

- 3.9 describe the manufacture of ethanol by passing ethene and steam over a phosphoric acid catalyst at a temperature of about 300°C and a pressure of about 60–70 atm
- 3.10 describe the manufacture of ethanol by the fermentation of sugars, for example glucose, at a temperature of about 30°C